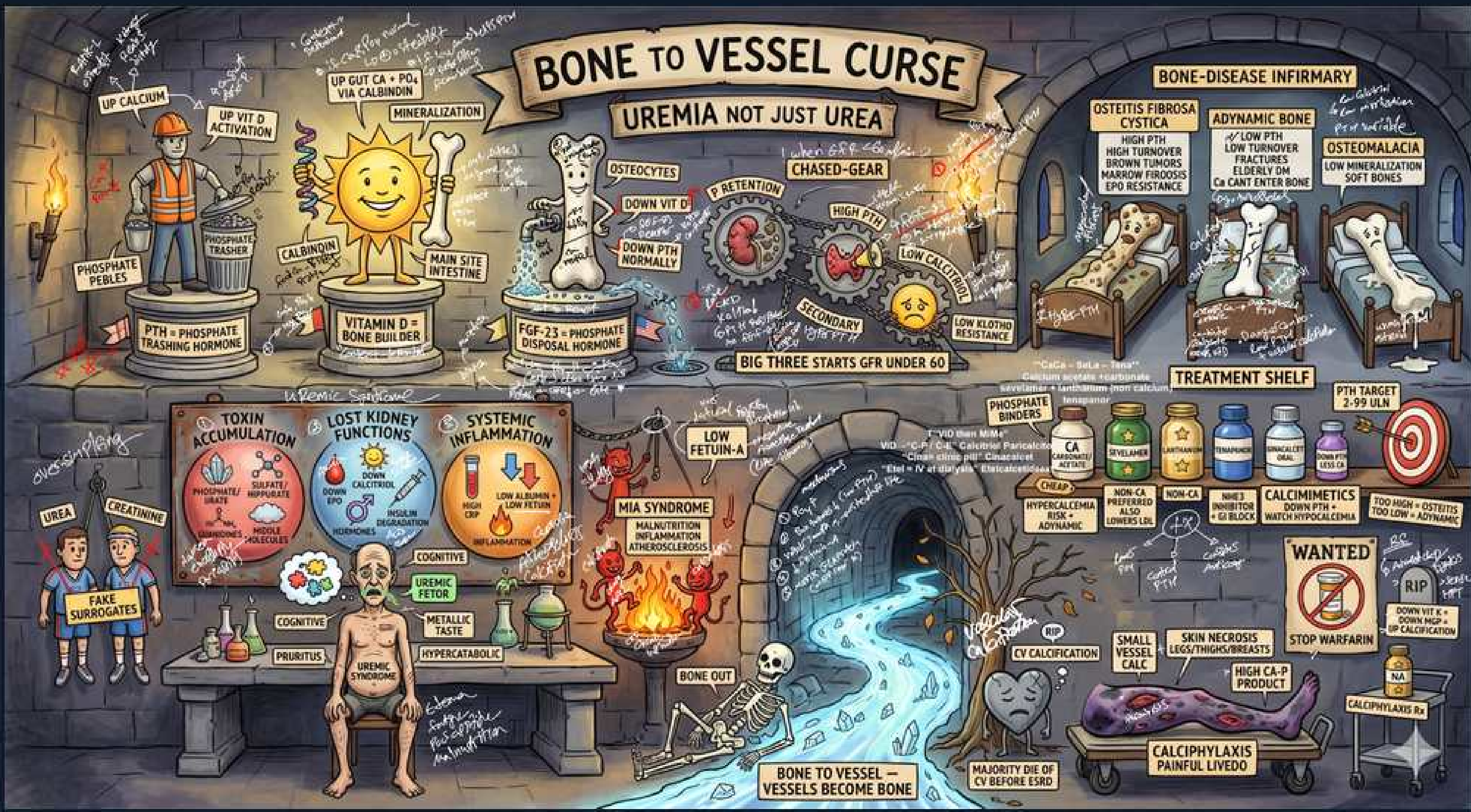
CKD — Bone, Phosphate & Calcium (CKD-MBD)



1. Bone-to-vessel curse banner

WHAT YOU SEE

Top banner 'Bone to vessel curse — uremia not just urea'.

WHAT IT ENCODES

CKD-Mineral and Bone Disorder (CKD-MBD) is a SYSTEMIC syndrome, not just a bone problem: it links three things — (1) abnormal labs (Ca, PO4, PTH, vitamin D, FGF-23), (2) abnormal bone (turnover/mineralization/volume = renal osteodystrophy), and (3) extraskeletal/vascular calcification. The 'curse' is that calcium is pulled OUT of bone and deposited INTO vessels, soft tissue, and valves. Mnemonic: the mineral that should build bone instead 'concretes the pipes.'

BOARD TRAP

Stem implies CKD bone disease is an isolated osteoporosis-type problem treated with a bisphosphonate — wrong answer. Discriminator: CKD-MBD is a triad (labs + bone + vascular calcification) and bisphosphonates are generally AVOIDED at low eGFR and can worsen adynamic (low-turnover) bone; you cannot treat the bone without addressing phosphate, PTH, and vitamin D together.

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2. High phosphate / phosphate peeks

WHAT YOU SEE

Worker labeled 'phosphate peeks', up arrow calcium/phosphate.

WHAT IT ENCODES

Failing kidneys cannot excrete phosphate, so PO4 is retained. Hyperphosphatemia is the central EARLY driver: it directly lowers ionized calcium, suppresses 1-alpha-hydroxylase (less calcitriol), and stimulates PTH and FGF-23 — and elevated calcium-phosphate product (>55–70) precipitates as metastatic/vascular calcification. Normal serum PO4 is ~2.5–4.5 mg/dL. Mnemonic: 'phosphate peaks first' and pulls the whole cascade along.

BOARD TRAP

Stem asks for the INITIATING event in secondary hyperparathyroidism of CKD — wrong answer is 'primary low calcium' or 'low vitamin D intake.' Discriminator: PHOSPHATE RETENTION is the inciting event (early, even before phosphate looks high because FGF-23 compensates); the hypocalcemia and low calcitriol are downstream. Treat by lowering phosphate first.

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3. Sun = low vitamin D

WHAT YOU SEE

Sun face labeled 'down vit D'.

WHAT IT ENCODES

The kidney's proximal-tubule 1-alpha-hydroxylase converts 25-OH vitamin D (calcidiol) to ACTIVE 1,25-(OH)2 vitamin D (calcitriol). In CKD this enzyme falls (suppressed further by high phosphate and FGF-23), so calcitriol drops → less gut calcium/phosphate absorption → hypocalcemia → more PTH. Treat with active analogs (calcitriol, paricalcitol, doxercalciferol), not just nutritional D, when PTH is rising. Mnemonic: the kidney is the 'final activation switch' for vitamin D.

BOARD TRAP

Stem: CKD patient with low calcitriol — wrong answer is to give high-dose plain cholecalciferol/ergocalciferol expecting it to fix the active-hormone deficit. Discriminator: the defect is in renal 1-alpha-HYDROXYLATION (activation), so an ACTIVE vitamin D analog (calcitriol/paricalcitol) is needed for the active-hormone deficiency; nutritional vitamin D only replaces substrate stores. Beware: active analogs RAISE calcium and phosphate — over-use causes hypercalcemia and adynamic bone.

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4. PTH → resistance (2° hyperPTH)

WHAT YOU SEE

Cascade caption in the upper-left flow reading 'PTH ↑ / phosphate / bone resistance' — the arrow into PTH shows the gland being driven up while bone turns deaf (resistant), the visual of secondary hyperparathyroidism.

WHAT IT ENCODES

Three stimuli drive SECONDARY hyperparathyroidism in CKD: high phosphate, low calcitriol, and low ionized calcium (plus parathyroid resistance to calcium/vitamin D via downregulated receptors). PTH rises to defend calcium by pulling it from bone (high turnover) — leading to osteitis fibrosa cystica. Persistent stimulation can become autonomous TERTIARY hyperparathyroidism (high PTH WITH high calcium). Mnemonic: 'Phosphate up, vitamin D down, Calcium down → PTH up.'

BOARD TRAP

Stem: long-standing CKD/dialysis patient now with HIGH PTH and HIGH calcium — wrong answer is to call it secondary hyperparathyroidism and just add a vitamin D analog. Discriminator: secondary hyperPTH has LOW/normal calcium; when the glands become autonomous you get TERTIARY hyperPTH (high PTH + HIGH calcium), which often needs a calcimimetic or parathyroidectomy, not more vitamin D (which would worsen hypercalcemia).

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5. FGF-23 / Klotho

WHAT YOU SEE

A small figure mid-scene labelled 'FGF-23' with a 'Klotho' note — drawn as the first alarm-raiser of the cascade, the earliest hormone to rise (with Klotho falling) before phosphate or PTH visibly climb.

WHAT IT ENCODES

FGF-23 (from osteocytes, acting with co-receptor Klotho) is the EARLIEST measurable change in CKD-MBD: it rises to promote renal phosphate excretion (phosphaturia) but also SUPPRESSES 1-alpha-hydroxylase, lowering calcitriol. Klotho falls early in CKD, causing FGF-23 resistance and ever-higher FGF-23 levels. High FGF-23 independently predicts LVH and mortality. Mnemonic: FGF-23 = the body's first (and ultimately failing) attempt to dump phosphate.

BOARD TRAP

Stem asks which mineral abnormality appears FIRST in early CKD — wrong answer is 'overt hyperphosphatemia' or 'high PTH.' Discriminator: FGF-23 ELEVATION (with falling Klotho) precedes overt phosphate rise and even precedes PTH elevation — it's the earliest CKD-MBD marker, keeping serum phosphate normal early at the cost of suppressing calcitriol.

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6. Osteitis fibrosa cystica

WHAT YOU SEE

Infirmary bed 'osteitis fibrosa cystica — high turnover'.

WHAT IT ENCODES

Osteitis fibrosa cystica = the classic HIGH-turnover renal osteodystrophy from severe, long-standing secondary hyperPTH. Excess PTH drives osteoclastic resorption → subperiosteal bone resorption (radial side of phalanges), 'salt-and-pepper' skull, brown tumors (osteoclast-rich lytic lesions), marrow fibrosis, and bone pain/fractures. PTH is markedly HIGH; alkaline phosphatase elevated. Mnemonic: 'too much PTH eats the bone hollow.'

BOARD TRAP

Stem: dialysis patient, very high PTH, bone pain, brown tumors, high bone turnover on biopsy — wrong answer is to over-suppress to a normal PTH or to confuse this with adynamic bone. Discriminator: HIGH PTH + HIGH turnover = osteitis fibrosa (treat by LOWERING PTH with binders/active D/calcimimetics); LOW PTH + LOW turnover = adynamic bone (the opposite end). They are managed in opposite directions.

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7. Adynamic bone disease

WHAT YOU SEE

Bed 'adynamic — low turnover'.

WHAT IT ENCODES

Adynamic bone disease = LOW-turnover osteodystrophy from OVER-suppressed PTH (too much calcium load, calcium-based binders, excess active vitamin D, or calcimimetics) — bone formation and resorption are both low. Bones are fragile (fracture risk) and cannot buffer calcium, so calcium goes to vessels (worsening vascular calcification and hypercalcemia). It is now the most common form in dialysis patients. Mnemonic: 'frozen bone' — nothing moving in or out.

BOARD TRAP

Stem: dialysis patient on calcium binders + active vitamin D now with LOW PTH, hypercalcemia, and fractures — wrong answer is to push MORE vitamin D/calcium to 'strengthen bone.' Discriminator: over-suppression of PTH causes adynamic (low-turnover) bone; the fix is to BACK OFF calcium/vitamin D/calcimimetics and allow PTH to rise modestly. A mildly elevated PTH is protective in dialysis — do not drive it to normal.

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8. Osteomalacia / soft bones

WHAT YOU SEE

Bed 'osteomalacia — soft bones'.

WHAT IT ENCODES

Osteomalacia = defective MINERALIZATION of osteoid → soft, weak bones with pseudofractures (Looser zones), bone pain, and proximal myopathy. In CKD it stems from severe vitamin D deficiency/low calcitriol and historically from ALUMINUM accumulation (old aluminum-based phosphate binders / aluminum-contaminated dialysate). Mnemonic: 'normal amount of bone matrix, but it never hardens.'

BOARD TRAP

Stem: older dialysis patient with bone pain, fractures, low turnover who was on aluminum-containing binders/antacids — wrong answer is to attribute it solely to hyperparathyroidism. Discriminator: ALUMINUM toxicity causes low-turnover osteomalacia (and microcytic anemia, encephalopathy/dementia) — treat by removing aluminum sources and chelating with deferoxamine; this is distinct from PTH-driven high-turnover disease.

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9. Big three: GFR under 60

WHAT YOU SEE

Caption 'big three starts GFR under 60'.

WHAT IT ENCODES

Once eGFR <60 (CKD G3a and beyond), begin monitoring the 'big three' — calcium, phosphate, and PTH — plus 25-OH vitamin D and alkaline phosphatase, with frequency increasing as CKD advances. Goals: keep phosphate toward the normal range, avoid hypercalcemia, and keep PTH from rising excessively. Mnemonic: 'sixty is the trigger to start watching Ca / PO4 / PTH.'

BOARD TRAP

Stem: stable eGFR 70 (G2) patient — wrong answer is to start routine CKD-MBD labs and binders. Discriminator: CKD-MBD monitoring and intervention generally begin at eGFR <60 (G3a); checking the 'big three' or treating phosphate in G1–G2 with preserved function is premature. Conversely, ignoring Ca/PO4/PTH once eGFR <60 is the opposite error.

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10. Treatment shelf — binders

WHAT YOU SEE

Shelf of bottles: phosphate binders (calcium-based vs non-calcium).

WHAT IT ENCODES

Phosphate control is step one: dietary phosphate restriction PLUS phosphate binders taken WITH meals to bind dietary PO4 in the gut. Calcium-based binders (calcium carbonate, calcium acetate) are cheap but add calcium load; non-calcium binders — sevelamer (also lowers LDL) and lanthanum carbonate — avoid calcium loading; ferric citrate/sucroferric also bind PO4 (ferric citrate adds iron). Mnemonic: binders only work if taken WITH food (no food, no phosphate to bind).

BOARD TRAP

Stem: dialysis patient with vascular calcification/hypercalcemia needing phosphate control — wrong answer is to escalate calcium carbonate. Discriminator: calcium-based binders add to total calcium load and promote VASCULAR CALCIFICATION/adynamic bone, so NON-calcium binders (sevelamer/lanthanum) are preferred in high-risk patients (hypercalcemia, vascular calcification, low PTH). Also: a binder given between meals does nothing for phosphate — must be with food.

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11. PTH target 2–9x ULN

WHAT YOU SEE

Target/bullseye 'PTH target 2–9x'.

WHAT IT ENCODES

In DIALYSIS-dependent CKD (G5D), KDIGO suggests keeping intact PTH roughly 2–9x the upper limit of normal — deliberately ABOVE normal because some PTH is needed to maintain bone turnover in an end-organ-resistant state. Trend matters more than a single value; rising or markedly high PTH prompts treatment escalation. Mnemonic: aim 'a few times normal,' not perfectly normal.

BOARD TRAP

Stem: dialysis patient with PTH at 2x ULN and a provider wants to push it into the normal range — wrong answer is to escalate vitamin D/calcimimetics to fully normalize PTH. Discriminator: normalizing PTH in dialysis patients risks ADYNAMIC bone; a mildly-to-moderately elevated PTH (target ~2–9x ULN) is adaptive and intentional. Over-treatment is as harmful as under-treatment here.

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12. Calcimimetics (cinacalcet)

WHAT YOU SEE

Bottle 'calcimimetics'.

WHAT IT ENCODES

Calcimimetics (cinacalcet oral; etelcalcetide IV with dialysis) activate the parathyroid calcium-sensing receptor, making the gland 'think' calcium is high → they LOWER PTH AND lower serum calcium and phosphate simultaneously. Especially useful in secondary hyperPTH with high/normal calcium (where vitamin D analogs would worsen hypercalcemia). Mnemonic: calcimimetic 'mimics calcium' to fool the parathyroid into shutting off.

BOARD TRAP

Stem: secondary hyperPTH with hypercalcemia is given a vitamin D analog and calcium rises further — wrong answer is to keep escalating the active vitamin D. Discriminator: calcimimetics LOWER calcium (and PTH), whereas active vitamin D analogs RAISE calcium; with high-calcium secondary hyperPTH, a calcimimetic is preferred. Watch the opposite hazard — cinacalcet can cause HYPOcalcemia (monitor calcium).

13. Calciphylaxis — wanted / no warfarin

WHAT YOU SEE

'Wanted' poster, skin necrosis, 'stop warfarin'.

WHAT IT ENCODES

Calciphylaxis (calcific uremic arteriolopathy) = calcification of small dermal arterioles → ischemia and exquisitely PAINFUL skin necrosis/ulcers (often livedo-like, proximal thighs/abdomen), high mortality from sepsis. Risk factors: ESRD/dialysis, high calcium-phosphate product, hyperparathyroidism, WARFARIN, obesity, female sex. Treat: stop warfarin, lower Ca/PO4 (non-calcium binders, cinacalcet, low-calcium dialysate), IV sodium thiosulfate, wound care. Mnemonic: 'wanted: warfarin' — it is a key culprit.

BOARD TRAP

Stem: dialysis patient on warfarin develops painful necrotic skin plaques — wrong answer is to continue warfarin (or treat it as ordinary cellulitis/vasculitis). Discriminator: STOP WARFARIN — it inhibits vitamin-K-dependent matrix Gla protein (the natural inhibitor of vascular calcification), worsening calciphylaxis; bridge to a non-warfarin anticoagulant if one is truly needed, lower the Ca×PO4 product, and add sodium thiosulfate.

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14. Bone One / vessels calcify skeleton

WHAT YOU SEE

Skeleton by river, fire, 'bone to vessel — vessels become bone'.

WHAT IT ENCODES

Chronic mineral derangement (high Ca×PO4 product, high PTH/FGF-23, low Klotho, adynamic bone that can't buffer calcium) transforms vascular smooth-muscle cells into an osteoblast-like phenotype → MEDIAL vascular calcification (Mönckeberg), arterial stiffening, valvular calcification, and LVH. This is a MAJOR contributor to the cardiovascular mortality that is the #1 killer in CKD. Mnemonic: 'the vessels literally turn to bone.'

BOARD TRAP

Stem links CKD mineral disease to outcomes and asks the main consequence of poorly controlled phosphate/calcium — wrong answer is 'fractures alone.' Discriminator: the dominant lethal consequence of vascular calcification in CKD-MBD is CARDIOVASCULAR (arterial stiffness, LVH, valvular disease) leading to CV death — the leading cause of mortality in CKD — not bone fracture per se. This ties CKD-MBD directly to CV mortality.
